



# simpleVIS user guide

simpleVIS is a **lightweight real-time lighting visualiser**. Import an MVR, take live levels over Art-Net, sACN, USB DMX or CITP, fly a camera around the rig, and see your programming in volumetric haze.

It is for seeing what a rig will do before you are in the room with it — checking looks and coverage against a real patch, without a visualiser licence and without modelling the venue by hand.

**Before you rely on this:** the import chain, DMX evaluation and protocol parsers are tested against real files and real packet layouts. The full 119-fixture MA Demostage renders at 60–70 fps, and the receive path is verified over real sockets.

**No real console and no DMX interface has driven it yet** — every packet so far has come from a test that wrote it, or from the built-in demo source. CITP has had first contact with a real peer (Capture 2026), but Capture is a DMX *consumer* like simpleVIS, so **levels and patch from a real console remain unverified**.

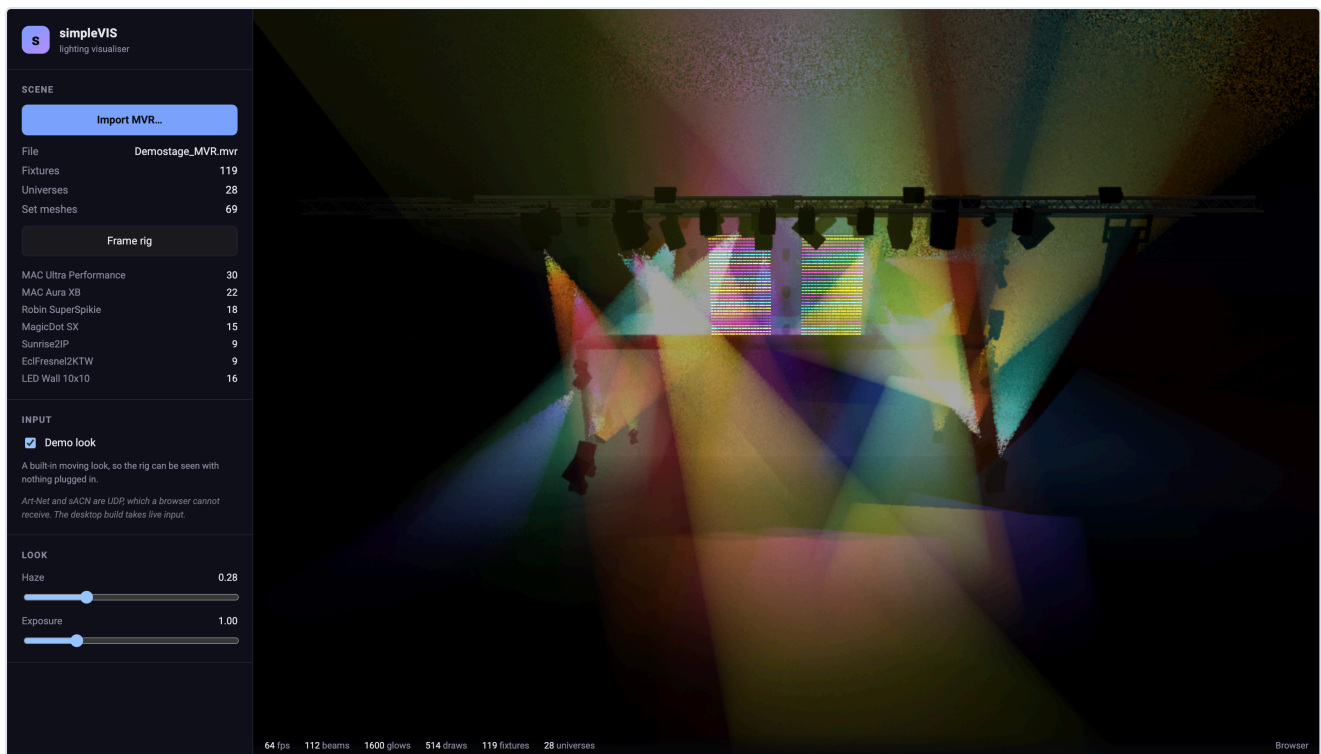
This project was built with AI assistance.

## Two builds, and why

**Art-Net and sACN are UDP, and a browser cannot receive UDP** — there is no flag, polyfill or workaround. So simpleVIS ships two targets from one codebase:

| Build                     | What it is                              | Live input                   |
|---------------------------|-----------------------------------------|------------------------------|
| <b>Desktop</b> (Tauri v2) | the real tool                           | Art-Net, sACN, USB DMX, CITP |
| <b>Hosted</b>             | offline previz and plot viewer in a tab | none — built-in demo look    |

The shared code is identical; only the transport differs. **The hosted build hides what it cannot do** rather than offering controls that fail.



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## Importing a rig

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**MVR** ( `.mvr` ), including the GDTF fixture definitions embedded inside it — geometry tree, DMX modes, beam angles, colour mixing.

**Import only. simpleVIS never writes MVR.** Nothing you do here can damage the file the desk gave you.

**Pixel-mapped fixtures work properly.** An MVR whose LED wall declares four template channels across a hundred `GeometryReference`s is resolved to its real **300-channel footprint, one DMX slot per pixel** — rather than being taken at face value as four channels.

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## Live input

| Protocol            | What's covered                                                                                           |
|---------------------|----------------------------------------------------------------------------------------------------------|
| <b>sACN (E1.31)</b> | Priority arbitration, CID source identity, multicast group joins per patched universe                    |
| <b>Art-Net 4</b>    | ArtDmx, and ArtPoll replies so consoles list simpleVIS as a node                                         |
| <b>USB DMX</b>      | <b>Enttec DMX USB Pro only.</b> An <i>Open</i> DMX USB has no receive path and cannot be an input at all |
| <b>CITP</b>         | Peer discovery (PINF), levels (SDMX), and fixture patch from the console (FPTC)                          |

**Two sources per universe are merged HTP or LTP with a 4-second timeout. A third is refused and reported** rather than silently dropped — so a mystery universe is a message, not a guess.

### What CITP does and doesn't cover

Three layers are implemented:

- **PINF** — peer discovery.
- **SDMX** — levels. It *does* carry them, as channel blocks; a common misconception says otherwise.
- **FPTC** — fixture patch, **so the patch can follow the desk without re-importing an MVR.**

**SXSr** is honoured, so a console that says "take my levels from Art-Net" is not *also* counted as a CITP source and double-merged.

**MSEX and CAEX — thumbnails and viewport streaming — are not implemented.** simpleVIS will not stream its viewport back to the desk.

## Troubleshooting

| Symptom                                          | Cause                                                                                                            |
|--------------------------------------------------|------------------------------------------------------------------------------------------------------------------|
| No live input options in the browser             | Correct — a browser cannot receive UDP. Use the desktop build.                                                   |
| USB DMX interface not receiving                  | An <i>Open</i> DMX USB has no receive path at all. Only the Enttec DMX USB Pro can be an input.                  |
| A universe won't take a third source             | By design — two sources merge, a third is refused and reported.                                                  |
| Levels stopped after a source disappeared        | The merge has a 4-second timeout before it drops a stale source.                                                 |
| LED wall renders as four channels, not per pixel | Shouldn't happen — <code>GeometryReference</code> expansion is tested. Worth reporting with the MVR.             |
| Console doesn't list simpleVIS as a node         | Art-Net discovery is ArtPoll/ArtPollReply; check the desk is polling the right subnet.                           |
| CITP peer found but no levels                    | The peer may be a consumer rather than a console — a visualiser sends neither levels nor patch.                  |
| Desk can't see simpleVIS's viewport              | Not implemented. MSEX/CAEX viewport streaming is out of scope for now.                                           |
| Fixture renders but doesn't move                 | Check the DMX mode the MVR declared matches what the desk is patched to — a mode is a template, not a footprint. |

## See also

- [AGENTS.md](#) — design notes, load-bearing invariants, and **the several ways an MVR will quietly render wrong**
- [README](#) — the two builds, protocol coverage and downloads

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<https://github.com/stoa-works-labs/simpleVIS>